B.Tech (ME)– IV-I Sem

23A03606	**AUTOMATION AND ROBOTICS (Professional Elective-V).**	**L**	**T**	**P**	**C**
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Course objectives: The objectives of the course are to	
1	**To introduce** the fundamentals of industrial automation, its types, components, and strategies across manufacturing systems.
2	**To explain** the working of automated flow lines and the methods used in assembly line balancing and flexible automation.
3	**To impart** knowledge about industrial robots, their configurations, anatomy, and applications in manufacturing processes.
4	**To develop** an understanding of manipulator kinematics, homogeneous transformations, actuators, and sensors used in robotics.
5	**To analyze** manipulator dynamics and apply trajectory planning techniques for robotic motion and obstacle avoidance.

Course Outcomes: On successful completion of the course, the student will be able to,		
1	Explain the need, types, and elements of automation systems and analyze different levels of automation strategies in industry.	L1,L2,L4
2	Demonstrate the operation of automated flow lines and apply methods for assembly line balancing and optimization.	L2,L3,L4
3	Identify robotic components and describe their configuration, degrees of freedom, and industrial applications in various processes.	L1,L2,L3
4	Apply transformation techniques and D-H notation to solve problems in robot kinematics and evaluate actuator and sensor selection.	L2,L3,L5
5	Analyze robot dynamics using Jacobians and Euler formulations and develop suitable trajectories for obstacle-free motion.	L3,L5,L6

UNIT-I

Introduction to Automation:

Introduction to Automation, Need, Types, Basic elements of an automated system, Manufacturing Industries, Types of production, Functions in manufacturing, Organization and information processing in manufacturing, Automation strategies and levels of automation, Hardware components for automation and process control, mechanical feeders, hoppers, orienters, high speed automatic insertion devices.

UNIT –II

Automated flow lines:

Automated flow lines, Part transfer methods and mechanisms, types of Flow lines, flow line with/without buffer storage, Quantitative analysis of flow lines. Assembly line balancing: Assembly process and systems assembly line, line balancing methods, ways of improving line balance, flexible assembly lines.

UNIT- III

Introduction to Industrial Robotics:

Introduction to Industrial Robotics, Classification of Robot Configurations, functional line diagram, degrees of freedom. Components common types of arms, joints grippers, factors to be considered in the design of grippers.

Robot actuators and Feedback components: Actuators, Pneumatic, Hydraulic actuators, Electric & Stepper motors, comparison. Position sensors - potentiometers, resolvers, encoders - velocity sensors, Tactile sensors, Proximity sensors.

UNIT- IV

Manipulator Kinematics:

Manipulator Kinematics, Homogenous transformations as applicable to rotation and transition - D-H notation, Forward inverse kinematics.

Manipulator Dynamics: Differential transformations, Jacobians, Lagrange - Euler and Newton – Euler formations. Trajectory Planning: Trajectory Planning and avoidance of obstacles path planning, skew motion, joint integrated motion - straight line motion.

UNIT- V

Robot Programming:

Robot Programming, Methods of programming - requirements and features of programming languages, software packages. Problems with programming languages.

Robot Application in Manufacturing: Material Transfer - Material handling, loading and unloading - Process - spot and continuous arc welding & spray painting - Assembly and Inspection.

Text Books:

1. Automation , Production systems and CIM,M.P. Groover /4thEdition, Pearson education (2016)
2. Industrial Robotics - M.P. Groover, TMH (1996)

References:

1. Robotics , Fu K S, McGraw Hill, 4th edition, 2010.
2. An Introduction to Robot Technology, P. Coiffet and M. Chaironze, Kogam Page Ltd. 1983 London.
3. Robotic Engineering , Richard D. Klafter, Prentice Hall
4. Robotics, Fundamental Concepts and analysis – Ashitave Ghosal ,Oxford Press, 1/e, 2006
5. Robotics and Control , Mittal R K &Nagrath I J , TMH.

Online Learning Resources:

Web References:

- http://www.cadcamfunda.com/cam_computer_aided_manufacturing
- http://wings.buffalo.edu/eng/mae/courses/460-564/Course-Notes/cnc- classnotes.pdf
- http://nptel.iitm.ac.in/courses.php?branch=Mechanical
- http://academicearth.org/courses/introduction-to-roboticsVideo references:-http://nptel.iitm.ac.in/video.php?courseId=1052